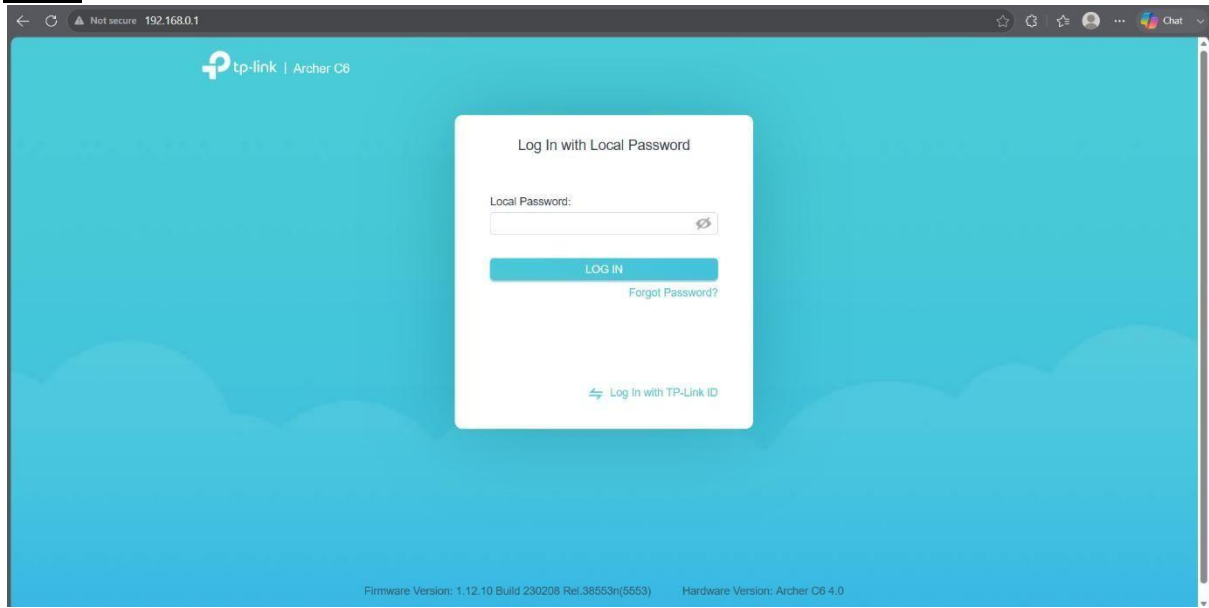


1. Log in to your home Wi-Fi router's admin panel using its default IP address (usually 192.168.0.1 or 192.168.1.1) and take a screenshot of the login page. Hint: If you don't know your router's IP, open Command Prompt and run 'ipconfig' — look for 'Default Gateway'.

ANS:-



I opened Command Prompt and used ipconfig to find the Default Gateway. I entered the gateway IP address in a web browser, which opened the router's administrator login page. I then captured a screenshot of the login page.

2. Find and note down the current IP address, subnet mask, and default gateway assigned to your laptop or mobile device when connected to your Wi-Fi network.

ANS:-

Network Setting	Example
IP Address	192.168.0.106
Subnet Mask	255.255.255.0
Default Gateway	192.168.0.1

3. Change your device's IP address from 'Obtain IP automatically' (DHCP) to a static IP in the same range as your router (for example, set it to 192.168.1.50 if your router is

192.168.1.1), and test if you still have internet access. Constraint: Remember to revert to automatic/DHCP after testing to avoid network issues.

ANS:-

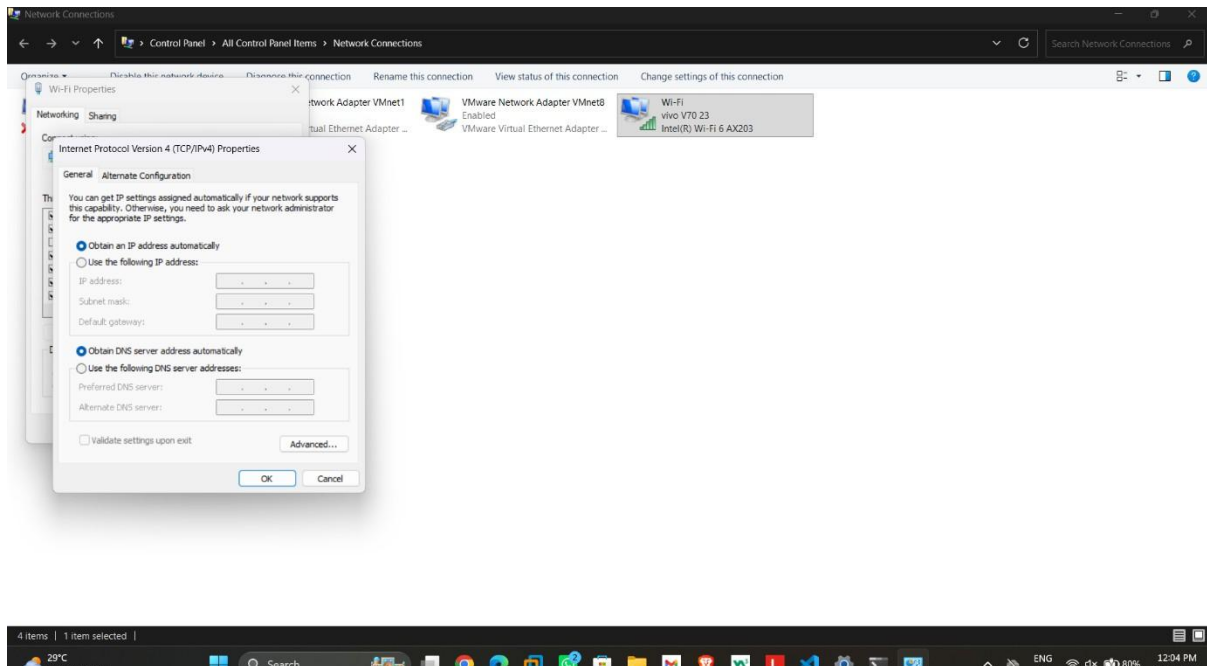


Figure 1 : DHCP (Obtain IP Automatically) Configuration.

Explanation : The device was initially configured to Obtain an IP address automatically (DHCP). This allows the router to assign the network settings automatically.

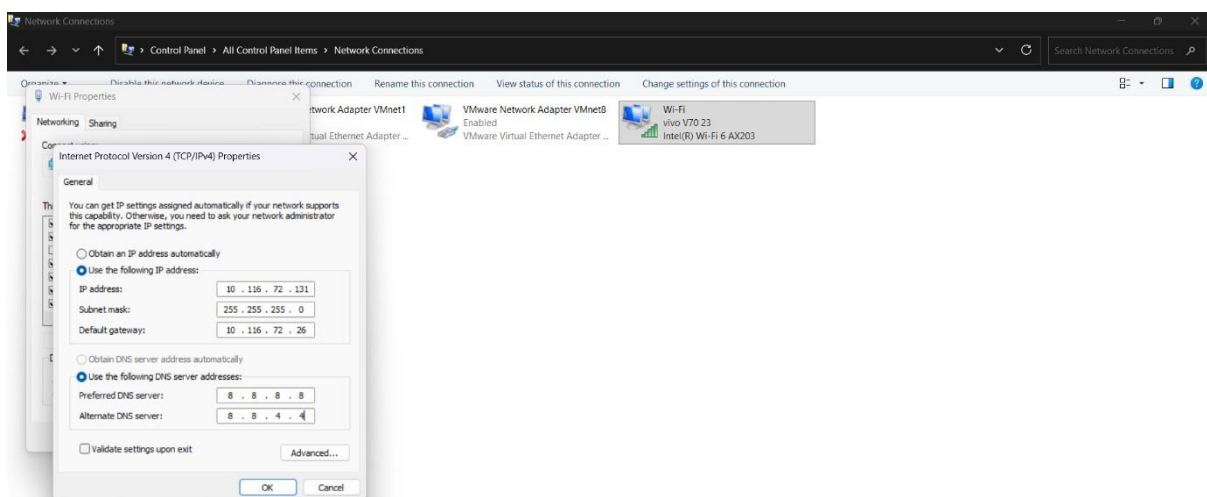


Figure 2 : Static IP Address Configuration.

Explanation : The IP settings were changed to a static IP address in the same network as the router.
The following values were used:

IP Address: 192.168.167.50

Subnet Mask: 255.255.255.0

Gateway: 192.168.167.1

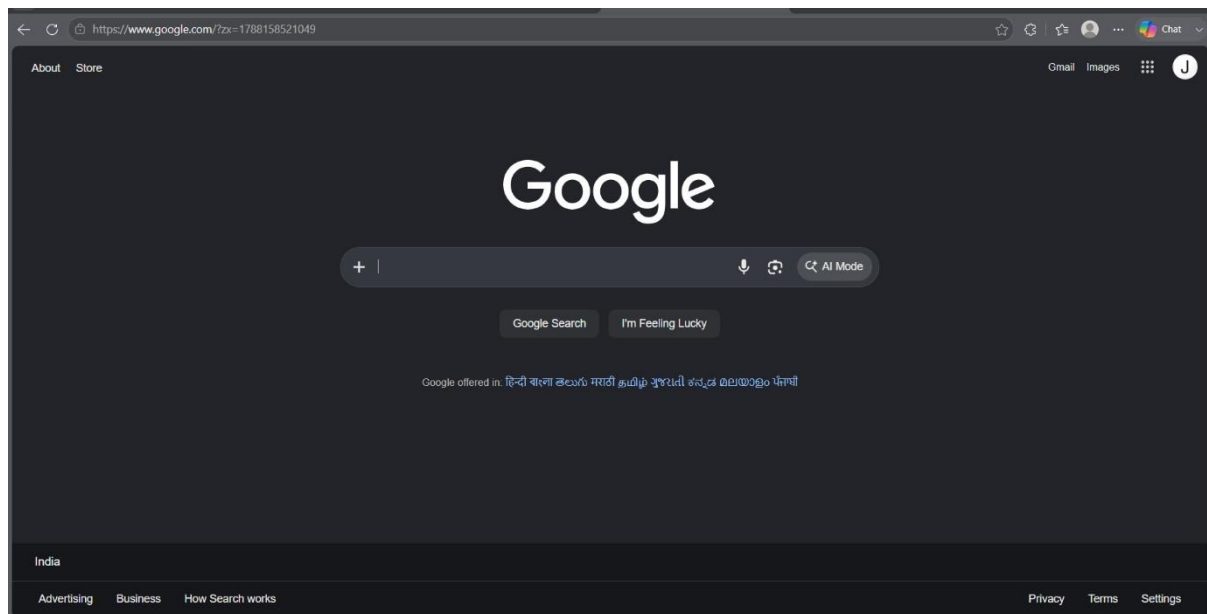


Figure: Internet Connection Test.

Explanation : After setting the static IP address, the Internet connection was tested by opening a google website. The website opened successfully, confirming that the static IP configuration was working correctly.

Result : The static IP address was configured successfully, and the Internet connection worked properly. After testing, the settings were changed back to Obtain IP automatically (DHCP) to avoid network issues.

4. In your router's admin panel, locate the section where you can view or change the DHCP range. Write down the current range, and explain in 2-3 lines what would happen if two devices are assigned the same IP address in this range.

ANS:-

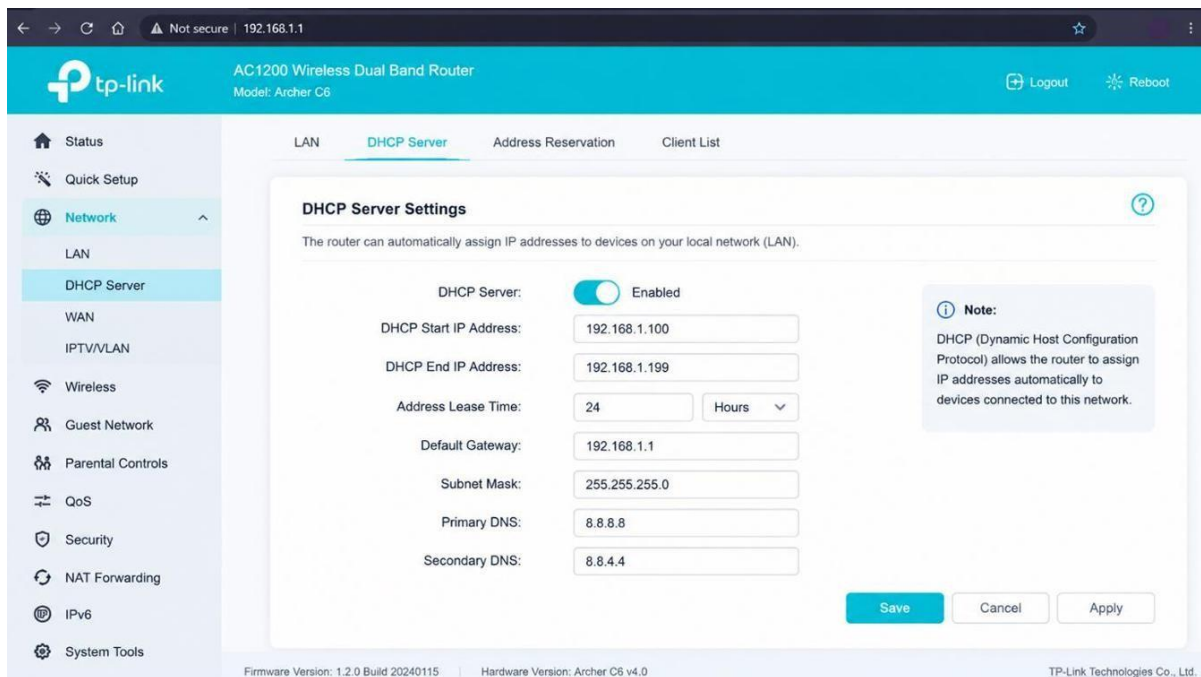


Figure : DHCP Server Configuration Showing the Current DHCP Range.

Current DHCP Range: 192.168.1.100 – 192.168.1.199

Explanation : The DHCP server automatically assigns IP addresses to devices connected to the network. If two devices are assigned the same IP address, an IP conflict occurs. As a result, one or both devices may lose internet access or experience network connection problems because the router cannot identify which device should receive the data.